3.5.1 Find the Square Root by Factorization Method

Example 3.19 Find the square root of the following expressions

(i) $256(x-a)^8(x-b)^4(x-c)^{16}(x-d)^{20}$ (ii) $\dfrac{144\ a^8\ b^{12}\ c^{16}}{81\ f^{12} g^4\ h^{14}}$

Solution (i) $\sqrt{256(x-a)^8(x-b)^4(x-c)^{16}(x-d)^{20}} = 16\left|(x-a)^4(x-b)^2(x-c)^8(x-d)^{10}\right|$

(ii) $\sqrt{\dfrac{144a^8b^{12}c^{16}}{81f^{12}g^4h^{14}}} = \dfrac{4}{3}\left|\dfrac{a^4b^6c^8}{f^6g^2h^7}\right|$

Example 3.20 Find the square root of the following expressions

(i) $16x^2+9y^2-24xy+24x-18y+9$ (ii) $(6x^2+x-1)(3x^2+2x-1)(2x^2+3x+1)$

(iii) $\left[\sqrt{15}x^2+\left(\sqrt{3}+\sqrt{10}\right)x+\sqrt{2}\right]\left[\sqrt{5}x^2+\left(2\sqrt{5}+1\right)x+2\right]\left[\sqrt{3}x^2+\left(\sqrt{2}+2\sqrt{3}\right)x+2\sqrt{2}\right]$

Solution (i) $\sqrt{16x^2+9y^2-24xy+24x-18y+9}$

$$= \sqrt{(4x)^2+(-3y)^2+(3)^2+2(4x)(-3y)+2(-3y)(3)+2(4x)(3)}$$

$$= \sqrt{(4x-3y+3)^2} = \left|4x-3y+3\right|$$

(ii) $\sqrt{(6x^2+x-1)(3x^2+2x-1)(2x^2+3x+1)}$

$$= \sqrt{(3x-1)(2x+1)(3x-1)(x+1)(2x+1)(x+1)} = \left|(3x-1)(2x+1)(x+1)\right|$$

(iii) First let us factorize the polynomials

$$\begin{aligned}
\sqrt{15}x^2+\left(\sqrt{3}+\sqrt{10}\right)x+\sqrt{2} &= \sqrt{15}x^2+\sqrt{3}x+\sqrt{10}x+\sqrt{2} \\
&= \sqrt{3}x\left(\sqrt{5}x+1\right)+\sqrt{2}\left(\sqrt{5}x+1\right) \\
&= \left(\sqrt{5}x+1\right)\times\left(\sqrt{3}x+\sqrt{2}\right) \\
\sqrt{5}x^2+\left(2\sqrt{5}+1\right)x+2 &= \sqrt{5}x^2+2\sqrt{5}x+x+2 \\
&= \sqrt{5}x(x+2)+1(x+2) = \left(\sqrt{5}x+1\right)(x+2) \\
\sqrt{3}x^2+\left(\sqrt{2}+2\sqrt{3}\right)x+2\sqrt{2} &= \sqrt{3}x^2+\sqrt{2}x+2\sqrt{3}x+2\sqrt{2} \\
&= x\left(\sqrt{3}x+\sqrt{2}\right)+2\left(\sqrt{3}x+\sqrt{2}\right) = \left(x+2\right)\left(\sqrt{3}x+\sqrt{2}\right)
\end{aligned}$$

Therefore,

$$\sqrt{\left[\sqrt{15}x^2+\left(\sqrt{3}+\sqrt{10}\right)x+\sqrt{2}\right]\left[\sqrt{5}x^2+\left(2\sqrt{5}+1\right)x+2\right]\left[\sqrt{3}x^2+\left(\sqrt{2}+2\sqrt{3}\right)x+2\sqrt{2}\right]}$$

$$= \sqrt{\left(\sqrt{5}x+1\right)\left(\sqrt{3}x+\sqrt{2}\right)\left(\sqrt{5}x+1\right)(x+2)\left(\sqrt{3}x+\sqrt{2}\right)(x+2)} = \left|\left(\sqrt{5}x+1\right)\left(\sqrt{3}x+\sqrt{2}\right)(x+2)\right|$$

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